JOURNAL OF SOCIAL COMPUTING
ISSN 2688-5255 01/06 pp297−308
Volume 2, Number 4, December 2021
DOI: 10.23919/JSC.2021.0028

Apologos: A Lightweight Design Method for Sociotechnical Inquiry

Luke Stark*

Abstract: While scholars involved in studying the ethics and politics flowing from digital information and communication systems have sought to impact the design and deployment of digital technologies, the fast pace and iterative tempo of technical development in these contexts, and the lack of structured engagement with sociotechnical questions, have been major barriers to ensuring values are considered explicitly in the R&D process. Here I introduce Apologos, a lightweight design methodology informed by the author's experience of the challenges and opportunities of interdisciplinary collaboration between computational and social sciences over a five-year period. Apologos, which is inspired by "design apologetics", is intended as an initial mechanism to introduce technologists to the process of considering how human values impact the digital design process.

Key words: values in design; values sensitive design (VSD); artificial intelligence; Values@Play; design methods; sociotechnical; ethics; values

1 Introduction

Human values pervade technical systems of all kinds[1], including computational technologies such as machine learning (ML) and other digital automation systems often termed artificial intelligence (AI)[2–4]. Over the past thirty years, work in fields such as science and technology studies (STS)[5, 6], social computing[7, 8], and critical studies of technology and race, gender, and sexuality[9–13] has interrogated the ways in which sociotechnical systems are conceived from and maintained in webs of normative preferences. Recent scholarship has paid particular attention to unpacking the granular technical affordances and design mechanisms in AI/ML[4, 14–19], through which particular human values are operationalized—and particular kinds of asymmetric power, injustice, and inequality maintained—in the everyday impacts of algorithmic technologies.

Alongside these academic developments, the increasingly obvious and deleterious societal impacts of social media platforms, artificial intelligence ventures, and other Silicon Valley firms have pushed the broad topic of "tech ethics", and the harm digital technologies do to marginalized groups, into international prominence[20–24]. Thanks to pressure from civil society groups, social justice organizations, activist scholars, and ordinary citizens, digital technology firms have been forced to begin to take responsibility for, and move to address their role in perpetuating and exacerbating social inequalities and power asymmetries.

As a result, digital technology firms have put much emphasis on high-level codes of ethical conduct around the development of technologies like AI systems, and have even begun to invest, albeit sporadically, in interdisciplinary teams of experts versed in the social impacts of computational media. However, methods and mechanisms to translate these high-level principles and diverse insights into actual decisions about products and systems are, on the whole, lacking[25]. Scholars involved in studying the ethics and politics of digital information and communication systems have long sought to have a concrete impact on the design and deployment of such artifacts in technical research and development (R&D) contexts such as academic laboratories and commercial

• Luke Stark is with the Faculty of Information and Media Studies, Western University, London, N6A 0A2, Canada. E-mail: cstark23@uwo.ca.
* To whom correspondence should be addressed.
Manuscript received: 2021-05-20; revised: 2021-11-20; accepted: 2021-11-25

© The author(s) 2021. The articles published in this open access journal are distributed under the terms of the Creative Commons Attribution 4.0 International License (http://creativecommons.org/licenses/by/4.0/).

development spaces[26–29]. However, the fast pace and iterative tempo of technical development in these contexts, and the importance of business decisions over wider societal concerns, have been major barriers to ensuring consideration of human norms and values is at play at every stage in the R&D process[30].

Urgent calls to grapple with the social, ethical, and normative implications of AI/ML and other computational systems on societies around the world make understanding and evaluating the ethics, norms, and values of all sorts of novel digital systems a necessity. Firms must move beyond lip-service to broader normative frameworks; critical and progressive responses to such technologies from lawmakers, regulators, civil society groups, and citizens in general can also benefit from such evaluations[21, 23, 31, 32].

Here, the author outlines a lightweight method for eliciting and evaluating ethics, norms, and human values in sociotechnical systems on a compressed time scale: Apologos. As a method, Apologos is inspired by the notion of "design apologetics"[33], which uses speculation to appraise technologies and their social impacts. This method seeks to present a coherent, practical, and principled approach to the problem of actively identifying norms, ethics, and values in a design process quickly. This method extends existing methodological frameworks[34–37] and draws on observations from a five-year case study of how conflicting human values intersect with the often complicated and contingent dynamics of designing digital systems[8].

The author also deploys insights from scholarship in design fiction[38], speculative design[39, 40], and the notion of "design apologetics"—thought experiments through which participants in a design process work backwards from existing or prospective artifacts to destabilize and make novel the normative notions behind digital technologies and systems. Such design apologetics ideally stimulate productive disorientation[41], and generative reflection about these technologies' possible social effects. This strategy is grounded in understanding and articulating how particular human norms, ethics, and values become incorporated in multifarious ways into the technical features of digital artifacts through design choices (both conscious and unconscious), and how such norms are made manifest in the use of technologies across diverse contexts[42].

As a method, Apologos seeks to use time itself as a visceral prompt to encourage novel thinking, and spark further in-depth reflection on and attention to the social contexts and lived realities of our everyday experience of technologies. Apologos is not intended as a panacea or replacement for more longitudinal or reflective design methods. As is a lightweight approach, it is potentially useful both in commercial settings and in broader participatory design contexts as an introduction to considering how human values are expressed in sociotechnical systems[43–45]. Apologos would be appropriate as an initial diagnostic exercise in a wide variety of digital design contexts: to begin to surface potentially confounding or complicated values tradeoffs[36]; point to spaces in the design process that might act as "values levers"[46]; and support space for novel engineering, participatory design, and refusal[47]. The theoretical and conceptual stakes of this article are thus twofold. First, what does the "nitty-gritty" of collaboration between computer scientists, social scientists, and humanists tell us about how to work across the socio-technical divide? Second, how can this experience shape an efficacious method for introducing audiences to the rich existing literature exploring how human values come to bear on the process of designing computational systems?

2 Lessons from Future Internet Architecture Project

Alongside already existing work on design prompts and methods such as Friedman's Valuse Sensitive Design[48] and Flanagan and Nissenbaum's Values@Play[36] methods, Apologos has been shaped by observations and insights drawn from the author's experience with the Future Internet Architecture (FIA) project, a multi-million dollar research project sponsored in two phases from 2010 to 2016 by the Computer and Network Systems (NETS) Division within the Directorate for Computer and Information Science and Engineering (CISE) of the National Science Foundation (NSF). The project involved four multi-institution research teams comprised primarily of computer scientists (henceforth referred to as the "computational teams"). These teams involved dozens of senior researchers and graduate students from more than fifteen different institutions. CISE also engaged several outside technical experts to

advise and interact with the projects on an ad-hoc basis. The whole initiative was led by David Clarke of the Massachusetts Institute of Technology (MIT), a pioneer in the current Internet's early network architecture[49–52].

The FIA technical teams were asked to "explore, design, and evaluate trustworthy future Internet architectures". Braman[53] notes, "decisions about technology design and network architecture are, today, de facto social policy." Recognizing this fact, the FIA Project also included the participation of the Values in Design Council, a group "funded by NSF to involve a set of social scientists, lawyers, and economists in the FIA design process". This group of experts, for which the author were a research assistant, included more than twenty leading scholars from information science, science and technology studies, technology law, and digital media studies. The FIA-VID collaboration was a rare example of a large, ongoing, and formal attempt to bring technical expertise together with sociocultural, legal, and policy insight in the service of computational design. As such, the lessons learned from both the successes and productive failures of these collaborations—which were multiple and nuanced—have already served as the basis for insights around how interdisciplinary teams can work collaboratively to design systems with human values in mind[8, 49].

This portion of the paper is thus grounded in qualitative materials, including collaborative and individual field notes, reports, and participant observation of more than a dozen FIA Principal Investigator meetings from 2010 through 2016. The experience of the Values in Design Council demonstrated the need to get interdisciplinary collaboration right between computer science and the social sciences/humanities. It also showed the related necessity, which became ever clearer over the course of the project, to develop a tool with which to introduce and familiarize computer scientists and engineers with the notion of engaging with human values in the design process quickly and efficaciously. As Shilton[8] observes in reference to her own experience as part of the FIA-VID project, such "interventions struggled to make values reflection consistently relevant and engaging" to members of the computational teams involved in the effort. These challenging elements of the FIA project experience have shaped Apologos as a method, and suggest avenues for further refinement and elaboration. What ontological and epistemological assumptions clash in the conversations between computer scientists and social scientists/humanists? What material practices and institutional or disciplinary norms help or hinder collaborations? And how do shared aims, desires, and values motivate or inhibit working together? Answers to these questions have shaped the development of Apologos as a method.

2.1 Key emergent theme: Clashing technical languages & epistemologies

Perhaps unsurprisingly, each academic discipline works with and within a particular technical language. While these technical languages are often related depending on the shared history and concepts of disciplines, the same terms in each discipline can suggest not just different technical definitions, but also imply radically different epistemologies (theories of professional knowledge) that require work to be made commensurate.

Throughout the course of the FIA-VID project, one of the chief practical obstacles to collaboration between computational and STS scholars was the way in which both linguistic definitions and epistemological priors were misunderstood by project participants, at least some (if not most) of the time. Shilton[8] observed one key definitional confusion in the context of FIA-VID was the status of the term "interoperable", which for the computational teams was implicitly synonymous for technologically "neutral". Shilton observes this definitional overlap prompted a key value assumption: "the asserted belief in the neutrality of architecture was at least partially an expression of a core value: the interoperability of infrastructure"[8]. Central to Shilton's observation is that definitions, and indeed their epistemological foundations, are grounded and guided by value judgments: in this case, the historical legacy of values consensus in network engineering has made end-to-end interoperability synonymous with technical and societal impartiality (ibid.).

Debates about the definitions and the broader epistemological meanings of terms such as privacy and security also exemplify how FIA-VID participants from the computational and STS teams struggled with developing common definitions. Two FIA Principal Investigator (PI) meetings focused their agendas on the

security and privacy implications of the various prototype architectures—the first, early in the project in 2011, and the second in 2015. The 2011 meeting also involved a group of outside technical security experts not attached to the four computational teams, who were tasked with advising those teams on their security and privacy plans as the projects moved forward from conceptualization to working prototypes. As privacy and security happened to be areas of considerable expertise for many Values in Design (VID) Council members, the meeting was also framed as an opportunity for the Council and technical teams to engage on an issue of common interest.

Unfortunately, it became apparent over the course of the meeting that "security" and "privacy" had different "technical" meanings for both groups. For the technical teams, "privacy" was generally articulated as a feature of their novel network architectures: something to be added as an additional layer or modification above more fundamental programming, but which was not in and of itself necessary for the functioning of the network. For the members of the VID Council, privacy was understood as a necessary outcome of network architecture, a default end state for ordinary users that ought to govern the technical teams' design decisions. One member of the VID Council noted that ordinary users of digital technologies rarely changed their default settings within their personal devices—and that privacy as a core human value ought to be integral to the teams' thinking.

In plain language, members of the technical teams wanted to know how to build systems and architectures that promoted user privacy as an actionable and imperative procedure, whereas members of the VID Council articulated privacy as an overarching end that could be enabled through a variety of different material, technological, and discursive means by the computational teams. The computational teams, a group made up primarily of computer scientists, operated primarily through what Abelson, Sussman, and Sussman term "procedural epistemology"—or in their words, "the study of the structure of knowledge from an imperative point of view, as opposed to the more declarative point of view taken by classical mathematical subjects." This epistemological logic, according to the authors, "provides a framework for dealing precisely with notions of 'how to'"[54]. This difference led to a focus on searching for concrete mechanisms through which to translate values into technical features. Differences between VID members and the computational teams thus did not seem to be grounded in a difference of underlying lived or embodied values per se. Rather, it was a disagreement around "lingo" tied to a deeper difference in epistemological constructions and prior assumptions held by the two groups.

In contrast, the members of the VID Council, as social scientists and humanists, operated through several overlapping epistemological frames. One, to borrow again from Abelson, Sussman, and Sussman, was a "declarative point of view"—which the authors associate with classical mathematical subjects, but which we here suggest is an epistemological frame that also fits with certain strands of humanistic and social scientific thought. Abelson, Sussman, and Sussman suggest that, "Mathematics provides a framework for dealing with precisely with notions of 'what is'." While the humanities and social sciences have never been accused of ontological clarity, their interest is often declarative or descriptive. Crucially, declarative epistemology is not the only flavor of knowledge construction in the human sciences—members of the VID Council also articulated discursive and critical epistemological frames through their comments.

In this particular instance of collaborative conversation, one that replayed itself around many other concepts across the life of the FIA project, both sides of the discussion were confused as to why a value each side agreed was valuable and important—privacy—seemed to nonetheless cause consternation (critically, in some contexts it became clear that privacy was not perceived as valuable by some in the conversation—a problem discussed later in the paper).

2.2 Key emergent theme: Conflicting disciplinary norms and incentives

Some of the most basic structural barriers to collaboration between the computational experts and VID Council members during the span of the FIA project involved the disparate and unaligned disciplinary incentives (and disincentives) around pursuing joint interdisciplinary projects. Studying interdisciplinary collaboration, particularly in the natural sciences, has become something of a cottage industry in recent years[55, 56]. However, as Callard and Fitzgerald[57] observe, this methodological focus has not extended to

other academic arenas: with very few exceptions[58], the authors note, "there has not yet been any significant emergence of research on practices of interdisciplinarity within the social sciences and humanities."[57] Worse, interdisciplinary collaborations between natural and social scientists, though increasingly vital to understanding and engaging with complex sociotechnical problems, are both rare in practice and understudied in terms of their collaborative dynamics[59].

Computer science as both an academic discipline and a set of professional practice is structured quite differently than the disciplines from which members of the VID Council were drawn (chiefly law, media and information studies, and the social sciences). Moreover, the specific structure of the FIA project also introduced structural incentives that discouraged efforts at interdisciplinary collaboration between the computational teams and VID Council. These challenges around interdisciplinary collaboration were exemplified by the relative physical and temporal separation of the VID Council from the four technical teams from the outset of the project. Over the course of the FIA project, VID Council members and members of the technical teams met at a series of semi-annual two-day PI meetings hosted by various member institutions. Because of the size of both the technical teams and of the VID Council, few if any project participants were present for every meeting. Because of the length of the project, there was considerable turnover among junior researchers (doctoral students and postdoctoral researchers) involved in the technical teams. And while members of each individual technical team were bound together both by the shared content of the project and by potentially a sense of competition *vis a vis* the other technical teams, there was more physical, social, and intellectual separation between the individual teams, and between the teams and the VID Council, across the life of the project. This segmentation kept project participants in both disciplinary and project-based silos, which made communication, let alone collaboration, an ongoing challenge.

These gaps did not go unnoticed by FIA Project organizers, who eventually sought to reduce the distance between FIA Council members and the technical teams. At the May 2012 PI meeting, the NSF announced that it would provide extra funding to "embed" members of the FIA Council within particular technical teams, and several VID Council members were ultimately affiliated to a greater or lesser extent with particular technical projects[8]. However, the default case for interactions remained large, structured presentations and Q&A sessions at PI meetings. While disciplinary segmentation is well-known challenge for interdisciplinary work, the experience of the FIA project highlighted an under-noted effect of these silos: the ways in which a lack of sociality and a high degree of emotional distance hindered the project's intellectual and scholarly goals. Previous scholarship has found that sustained social bonds and shared physical space are helpful in supporting interdisciplinary inquiry by connecting individuals interpersonally as well as intellectually[60].

As Callard et al.[60] note, hierarchies within interdisciplinary collaborations can often short-circuit sustained interactions. Given the intermittent nature of most interactions between Council participants and technical team members, it is unsurprising that relatively few social bonds formed over the course of the project between the two groups. Divergent publishing conventions and expected project outputs and timelines also posed challenges for sustained cross-disciplinary collaboration across the life of the FIA project. For the computer scientists involved in the project, the core technical problems around the design of network architecture required a different collaborative cadence and pace of publication than studies by the social scientists in the VID Council. Interdisciplinary collaborations were curtailed as much by divergent professional schedules and incentives as they were by a lack of social connection between VID Council members and members of technical teams.

3 Apologos as Design Prompt

One important lesson of the FIA project is that many technical experts have little vocabulary or formal training in engaging with sociotechnical questions, but are eager and excited to do so if supported by appropriate research and design methods. One strategy that the Values in Design Council deployed with some success over the course of the FIA project was around design scenarios: suggesting the computational teams sketch out how their proposed systems might work in real-world conditions and what such conditions indicated about the values of the systems at hand.① Apologos is first and foremost inspired by this

① Particular credit goes to VID Council members James Grimmelmann and Chris Hoofnagle for their initial championing of this approach.

experience, which could nonetheless have perhaps been improved by a standardized, lightweight set of prompts to help initiate such discussions across both the Council and computational groups.

As noted above, Apologos draws from the rich body of extant scholarship and design practice on values and human design, including how human values can be elicited or translated into technical means through particular design prompts or toolkits. Structured design prompts have become a popular method for facilitating design research and participatory co-design over the past decade in critical HCI and related fields. These prompts often feature a toolkit of playing cards or make use of other gamic elements as mechanisms to structure and vary deign outcomes[61]. Popular examples of such card-based design prompts include the *Envisioning Cards* from the University of Washington's Value-Sensitive Design (VSD) group[34, 62] among many others[63]. More broadly, the notion of "design sprints" and similar time-constrained methods to facilitate design work has proliferated in both academia and the digital technology sector[64]. Recent frameworks and toolkits for applying VSD methods to contemporary tech development environments are a salutary means of effectively integrating these design traditions to an applied context[65].

Apologos is inspired by these existing design traditions and methods. It also draws on, and is named after, the notion of "design apologetics", proposed by interface designers Nathan Shedroff and Christoper Noessel in *Make It So: Interaction Design Lessons from Science Fiction* (2012). Shedroff and Noessel borrow the term "apologetics" from theology, where it refers to reasoned argument justifying a religious doctrine. In the context of design in science fiction films and television programs, Shedroff and Noessel note they searched for ways fictional technologies "could" be explained to work, which "led to some interesting insights about the way technology should work"[33]. Other recent academic work has centered using speculative fiction as a prompt to produce novel design insights around human values and more broadly as a tool for critical design practice in computational settings[38, 66, 67].

Building on this prior work, Apologos is intended as an intervention to enable interdisciplinary groups to concisely consider a technology's sociotechnical impacts using design apologetics (as outlined in Table 1). An Apologos session is essentially a highly compressed version of the three-stage process laid out in Refs. [35, 36]: discovery of relevant values in a particular situation, implementation of those values as technical features, and verification that the assumptions made in these first two steps are broadly pertinent. Ideally, Apologos should be deployed either at the very outset of a particular design project or as a pedagogical prompt to introduce new audiences to sociotechnical analysis. Each session should be relatively brief: no less than 1 h, and potentially two hours or slightly more. The exercise is most practically tractable if undertaken by groups of 3−4 individuals; it is beneficial for larger design teams to break up into these smaller cadres and then reassemble as a larger unit for the final phase of the initial exercise as per below.

The first step in an Apologos session is for the session facilitator to ground participants in two definitions. First, ethos: a moral habit, character, disposition, or custom.

Table 1 Apologos summary.

Phase	Component (total 60 min)
Discovery	Definitions (8 min)
	Brainstorming (2 min)
	List development (5 min)
Implementation	Design apologetics (2 min)
	Apologetics application (8 min)
	(Break for 5 min)
	Value judgment (5 min)
	Re-design (5 min)
Evaluation & follow-up	Reflection (5 min)
	Sharing across groups (8 min)
	Exercise feedback (7 min)
	Method transition (indeterminate)

Second, techne: a variable and context-dependent art or craft **(approximately 8 min, in a 1 h session)**. Attentive readers will note here how the notion of ethos stands in for the broader definition of the "social" common to science and technology studies (STS) literature on sociotechnical systems. Any ethos is inherently a communal, and thus social, undertaking, but the term's normative connotation, and its emphasis on habituation, make it particularly apt for this exercise. After laying out these definitions, the facilitator should note the significant practical overlap between these two definitions around usual custom and lived concreteness—by extension the ways particular sociotechnical systems are inherently ones with unique sets of norms and values.

The participating small groups should then brainstorm three to four examples of everyday situations, scenarios, or activities where the definitions of ethics and technics already provided might be at play: a sociotechnical situation involving both aspects **(2 min in a 1 h session/ time elapsed: 10 min)**. Once participants have identified a short list of scenarios or situations, such as flagging down a self-driving ride share vehicle or having one's mobile devices searched at a national border, each group should collectively develop, for just one of those scenarios, two parallel lists: one of 3−4 ethical principles or values the group associates with the situation, and one of 3−4 technical/material elements or features of the situation **(5 min in a 1 h session/time elapsed: 15 min)**. The facilitator should encourage these lists to be in parallel vertical columns on one sheet of paper. As a further prompt, the facilitator can also emphasize the utility of considering what Flanagan and Nissenbaum term "values seams"[36], or "places where multiple values are held in tension" within particular technologies or technical systems, as a means to remember that no technology is in any way "neutral".

The initial steps of an Apologos session—highlighting both the difference and potential overlaps in the definitions of *techne* and *ethos* and grounding this general discussion in particular domain contexts—are intended to immediately highlight the processual versus declarative divide in considering a value such as privacy in particular sociotechnical milieus. The group discussion intended to support these determination serves as what Shilton terms a "values lever"[46], or "practices that pry open discussions about values in design and help the team build consensus around social values as design criteria". Values levers are valuable to delicately de-lace Flanagan and Nissenbaum's "value seams", enabling collaborative examination and discussion of the multiple value perspectives within a particular sociotechnical apparatus. Through these steps, definitional and even epistemological differences can be, if not resolved, at least surfaced and recognized as such.

When these first stages of the exercise are complete, the facilitator should introduce the notion of design apologetics **(2 min in a 1 h session/time elapsed: 17 min)** to participants (as a reminder, apologetics are reasoned arguments or writings in justification of something, typically a theory or religious doctrine). Groups should not be informed about the details of the design apologetics stage before they formulate their two initial lists.

In their existing groups, participants should then perform apologetics across their lists **(8 min in a 1 h session/time elapsed: 25 min)**, speculating about or imagining reasonable ways a designer might pair the ethical principles or values discerned by the group in the first phase of the exercise with the technical features the group has picked, inasmuch as technical features can express a value or make it concrete. It is critical that participants do not change the content of either list to make this exercise easier or tidier. If members of the group cannot draw a reasonable connection between the values they first identified and the initial material or technical features, this failure should be specifically noted as another example of Flanagan and Nissenbaum's notion of "values seams". It is to encourage the exposure of such values seams that groups should not be informed about the design apologetics stage before they formulate their initial lists. This stage should be immediately followed by a stretch break for decompression and informal conversation about the exercise **(5 min in a 1 h session/time elapsed: 30 min)**.

After the break, the facilitator should focus participants on the most discordant or least convincing pairing of principles and technical features out of their list. Group members should then decide together whether they judge, in the context of the other values and features identified, whether it is the ethical value or the technical feature that is more important to the broader goals of the project, situation, or milieu **(5 min in a 1 h**

session/time elapsed: 35 min). For groups with little design experience, the facilitator might note that in making such a judgment, participants are expressing their own values as designers. For groups with more experience with sociotechnical analysis, the facilitator can observe beforehand that how each group understands and bounds their project or situation is itself a values judgment that inevitably shapes the decisions in this step.

After each group has made a judgment collectively, the facilitator should encourage group members to either (1) brainstorm a new technical feature that better makes concrete the value or principal the group has decided to prioritize, or (2) identify another principal or value (even a "negative" or unwanted one) suggested by the technical feature being prioritized to replace the initial ill-matched value (**5 min in a 1 h session/time elapsed: 40 min**). Group members should then record answers to the following questions collectively (**5 min in a 1 h session/time elapsed: 45 min**): How might this planned change interact with the other values and technical features already identified in the scenario or milieu? How might this change affect the broader parameters of the scenario or situation originally laid out?

Finally, the facilitator should quickly organize a jigsaw in which each small group splits and new groups of 2−3 people are constituted; if there is only one small group, then the following two steps can be combined. Each group member should briefly present the work of their original group to their new group mates and ask them to imagine themselves as people in the scenario the original group explored **(8 min in a 1 h session/time elapsed: 53 min).** Participants should ask their new groupmates whether there are values or technical elements of the original group's assessment with which they disagree, or that they would add or take away. Moreover, each member should ask the others how the change made by the original group around either a value or technical feature would affect them as imagined subjects in the scenario. For the last portion of the hour span (**7 min in a 1 h session/time elapsed: 60 min**), participants should reconvene into a large group and the facilitator should ask for feedback on the design exercise, including about what was satisfying, enlightening, or useful; what was unsatisfying, frustrating, or incomplete; and if appropriate, what changes would improve the exercise for participants in a particular context.

One benefit to a lightweight design exercise like Apologos is that it can be deployed expeditiously and readily understood by teams from diverse disciplines and backgrounds. However, Apologos should by no means be the final step in a design and development process: as a final step, the group should make a plan to transition to a more fully developed method for designing with values in mind as appropriate. Frameworks such as Value Sensitive Design (VSD)[34, 48, 65], worth-centered design[68, 69], reflective design[70], adversarial design[71], and Values@ Play[35, 36], and working with values hypotheses[42] are all potential options for such a framework.

As an exercise, Apologos was designed by considering the experience of FIA participants, who faced a related set of conflicting norms grounded in the disciplinary and institutional nature of large-scale academic research. Insights from these successes and failures point to ways design exercises like Apologos can help bridge these institutional and professional gaps and to moments where broader and longer-term work is needed above and beyond any individual design technique. With both social and disciplinary challenges in mind, Apologos is designed to be a group activity that ensures a large group of disparate experts can participate in discussions around values and technologies and to be sufficiently brief that it can be deployed even at one- or two-day meetings. The social element of Apologos provides one possible scaffold for broader collaboration, or at least engagement, between members of diffuse interdisciplinary teams. Given sustained interactions and shared physical proximity are clearly ideal in this regard[60], Apologos aims to begin the process of social mixing, while building shared incentives around speculative design.

4 Conclusion: Disagreeing Over Values

Here, the author has proposed a lightweight design method, Apologos, intended to elicit and evaluate ethics, norms, and human values in sociotechnical systems on a compressed time scale. As already noted, Apologos is not a panacea or replacement for more comprehensive design methods or structural mechanisms to facilitate both interdisciplinary collaboration and broad, truly participatory responses to sociotechnical problems[72]. The author notes two particular limitations: the composition of the participants in an Apologos session,

and the potentially limited impact of such a speculative design exercise.

As with all design, the possible outputs of an Apologos session will be bounded by the experiences, positionality, and perspectives of those in the room—as well as the shortcomings inherent in the ideology of "design" thinking itself[73]. This is of course an argument for engaging diverse teams of participants in the work of design[19]. However, it is also a warning that, as should be obvious, no single Apologos session is sufficient to adequately explore the sociotechnical terrain of any given computational artifact. The broader literature on the merits and challenges of participatory design in computing and digital media points to the shortcomings of even comprehensive methods for widespread inclusion in the design process, and Apologos should thus be understood as an introduction to this much larger and thornier area of practice. Apologos is an entry point into sociotechnical/interdisciplinary collaboration but sustaining and fostering that collaboration require much additional work.

A second limitation, related to the first, concerns how impactful the outcomes of one, or even many Apologos sessions can realistically be in changing the usual activities of a startup, corporation, or public institution[21, 74]. An Apologos session, or even several, will provide sufficient insight to change policy. However, as an entry to thinking critically about the sociotechnical landscape, Apologos is potentially useful in providing a language and framework for technologists and others who have not had a structured means to consider such questions before. Apologos should be seen thus as an introductory component of a much broader set of developments around the training, education, regulation, and design of digital technologies.

To conclude, the author wants to flag one challenge that can be adequately solved only through such broader structural mechanisms: the way divergent disciplinary, professional, and personal norms and expectations fundamentally shape how values are articulated and incorporated into design decisions, and how such disagreements can reflect a fundamental clash of values[42]. Although differences in social, professional, and epistemological norms comprised many of the roadblocks to collaboration during the course of the FIA project, at times project participants disagreed on an ontological level: about the fundamental values at issue in the development of network architecture and about which values are more or less important. Privacy is one clear example: some members of the computational teams were simply not convinced that human privacy as a value superseded others such as speed, user convenience, or network security.

These disagreements entail what Flanagan and Nissenbaum term "values trade-offs", or moments within the process of creation in which some values are prioritized over others[36]. While Apologos provides a mechanism for surfacing and highlighting such values trade-offs and their possible effects, it does not provide guidance *per se* on how to adjudicate between such tradeoffs. As Flanagan and Nissenbaum observe, "It is not surprising to find that design projects (particularly those with multiple requirements, goals, constituencies, and constraints) are rife with clashes and conflicts"[36]. And while Apologos does provide for identifying "negative" effects of potential design decisions as a means for some guidance, on what scale such negatives are judged remains at the discretion of the particular group of people doing the designing. Their clashes and conflicts, and the broader structural conflicts they represent, will be both unique to each situation and challenging in all cases[19, 42].

Nonetheless, Apologos has utility as one method among many in the broader conversation around how to account for human values and ethics in digital technologies. Lightweight exploratory methods like Apologos will ideally support space for broader conversations around novel engineering solutions[38], the necessity of participatory design across technical fields[25], and the necessity of refusal or non-deployment as an R&D option[47]. As sociotechnical analyses of pressing societal challenges become more urgent, and more frequent, we need ongoing focus on the work of interdisciplinary translation and design implementation as we navigate the ethics and values designed into and emerging from digital technologies.

References

[1] L. Winner, Do artifacts have politics? in *The Whale and the Reactor: A Search for Limits in an Age of High Technology*, L. Winner, ed. Chicago, IL, USA: University of Chicago Press, 1986, pp. 19–39.

[2] B. Friedman and H. Nissenbaum, Bias in computer systems, *ACM Transactions on Information Systems*, vol. 14, no. 3, pp. 330–347, 1996.

[3] C. O'Neil, *Weapons of Math Destruction*. New York, NY,

USA: Broadway Books, 2017.
[4] A. Selbst, S. Friedler, D. Boyd, S. Venkatasubramanian, and J. Vertesi, Fairness and abstraction in sociotechnical systems, in *Proc. the Conference on Fairness, Accountability, and Transparency*, Atlanta, GA, USA, 2019, pp. 59–68.
[5] B. Pfaffenberger, "If I want it, it's OK": Usenet and the (outer) limits of free speech, *The Information Society*, vol. 12, no. 4, pp. 365–386, 1996.
[6] L. Suchman, J. Blomberg, J. E. Orr, and R. Trigg, Reconstructing technologies as social practice, *American Behavioral Scientist*, vol. 43, no. 3, pp. 392–408, 1999.
[7] B. Friedman, P. H. Kahn Jr., J. Hagman, R. L. Severson, and B. Gill, The watcher and the watched: Social judgments about privacy in a public place, *Human-Computer Interaction*, vol. 21, no. 2, pp. 235–272, 2006.
[8] K. Shilton, Engaging values despite neutrality, *Science, Technology, & Human Values*, vol. 43, no. 2, pp. 247–269, 2018.
[9] J. Gaboury, Becoming NULL: Queer relations in the excluded middle, *Women & Performance*: *A Journal of Feminist Theory*, vol. 28, no. 2, pp. 143–158, 2018.
[10] M. K. Scheuerman, E. Denton, and A. Hanna, Do datasets have politics? Disciplinary values in computer vision dataset development, *Proceedings of the ACM on Human-Computer Interaction*, vol. 5, no. CSCW2, pp. 1–37, 2021.
[11] R. Benjamin, Assessing risk, automating racism, *Science*, vol. 366, no. 6464, pp. 421–422, 2019.
[12] S. U. Noble, *Algorithms of Oppression: How Search Engines Reinforce Racism*. New York, NY, USA: New York University Press, 2018.
[13] J. Buolamwini and T. Gebru, Gender shades: Intersectional accuracy disparities in commercial gender classification, in *Proc. the 1st Conference on Fairness, Accountability and Transparency*, New York, NY, USA, 2018, pp. 77–91.
[14] S. Barocas and A. D. Selbst, Big data's disparate impact, *California Law Review*, vol. 104, pp. 671–732, 2016.
[15] A. L. Hoffmann, Data violence and how bad engineering choices can damage society, Medium, https://medium.com/s/story/data-violence-and-how-bad-engineering-choices-can-damage-society-39e44150e1d4 , 2018.
[16] A. L. Hoffmann and L. Stark, Hard feelings — inside out, Silicon Valley, and why technologizing emotion and memory is a dangerous idea, https://lareviewofbooks.org/essay/hard-feelings-inside-out-silicon-valley-and-why-technologizing-emotion-and-memory-is-a-dangerous-idea, 2015.
[17] M. M. Malik, A hierarchy of limitations in machine learning, https://arxiv.org/pdf/2002.05193.pdf, 2020.
[18] M. Raghavan, S. Barocas, J. Kleinberg, and K. Levy, Mitigating bias in algorithmic employment screening: Evaluating claims and practices, arXiv preprint arXiv: 1906.09208, 2019.
[19] B. Fish and L. Stark, Reflexive design for fairness and other human values in formal models, in *Proc. 2021 AAAI/ACM Conference on AI, Ethics and Society*, Virtual Event, USA, 2021, pp. 89–99.
[20] V. Eubanks, *Automating Inequality: How High-Tech Tools Profile, Police, and Punish the Poor*. New York, NY, USA: St. Martin's Press, 2018.
[21] D. Greene, A. L. Hoffmann, and L. Stark, Better, nicer, clearer, fairer: A critical assessment of the movement for ethical artificial intelligence and machine learning, https://hdl.handle.net/10125/59651, 2019.
[22] A. Rességuier and R. Rodrigues, AI ethics should not remain toothless! A call to bring back the teeth of ethics, *Big Data & Society*, vol. 7, no. 2, p. 205395172094254, 2020.
[23] S. Mohamed, M. -T. Png, and W. Isaac, Decolonial AI: Decolonial theory as sociotechnical foresight in artificial intelligence, *Philosophy & Technology*, vol. 33, pp. 659–684, 2020.
[24] B. Green, The contestation of tech ethics: A sociotechnical approach to technology ethics in practice, *Journal of Social Computing*, doi: 10.23919/JSC.2021.0018.
[25] M. A. Madaio, L. Stark, J. W. Vaughan, and H. Wallach, Co-designing checklists to understand organizational challenges and opportunities around fairness in AI, in *Proc. the 2020 CHI Conference on Human Factors in Computing System*, Honolulu, HI, USA, 2020, pp. 1–14.
[26] N. Manders-Huits and M. Zimmer, Values and pragmatic action: The challenges of introducing ethical intelligence in technical design communities, *International Review of Information Ethics*, pp. 1–8, 2009.
[27] I. Poel, An ethical framework for evaluating experimental technology, *Science and Engineering Ethics*, vol. 22, no. 3, pp. 667–686, 2015.
[28] J. van den Hoven, P. E. Vermaas, and I. van de Poel, Design for values: An introduction, in *Handbook of Ethics, Values, and Technological Design*, J. van den Hoven, P. E. Vermaas, and I. van de Poel, eds. Dordrecht, the Netherlands: Springer, 2015, pp. 1–7.
[29] I. van de Poel, J. N. Fahlquist, N. Doorn, S. Zwart, and L. Royakkers, The problem of many hands: Climate change as an example, *Science and Engineering Ethics*, vol. 18, no. 1, pp. 49–67, 2011.
[30] S. Gürses and J. van Hoboken, Privacy after the agile turn, in *Cambridge Handbook of Consumer Privacy*, J. Polonetsky, O. Tene, and E. Selinger, eds. Cambridge, UK: Cambridge University Press, 2018, pp. 579–601.
[31] E. Donahoe and M. M. Metzger, Artificial intelligence and human rights, *Journal of Democracy*, vol. 30, no. 2, pp. 115–126, 2019.
[32] Y. Stevens and A. Brandusescu, Weak privacy, weak procurement: The state of facial recognition in Canada, *SSRN Electron J*, doi: 10.2139/ssrn.3857355.
[33] N. Shedroff and C. Noessel, *Make It So: Interaction Design Lessons from Science Fiction*. New York, NY, USA: Rosenfeld Books, 2012.
[34] B. Friedman, D. G. Hendry, and A. Borning, A survey of value sensitive design methods, *Foundations and Trends® in Human–Computer Interaction*, vol. 11, no. 2, pp. 63–125, 2017.
[35] M. Flanagan, D. C. Howe, and H. Nissenbaum, Embodying values in technology: Theory and practice, in *Information Technology and Moral Philosophy*, J. van den Hoven and J. Weckert, eds. Cambridge, UK: Cambridge University Press, 2008, pp. 322–353.
[36] M. Flanagan and H. Nissenbaum, *Values at Play in Digital*

Games. Cambridge, MA, USA: The MIT Press, 2014.
[37] K. Shilton, Values and ethics in human-computer interaction, *Foundations and Trends® in Human–Computer Interaction*, vol. 12, no. 2, pp. 107–171, 2018.
[38] R. Y. Wong and D. K. Mulligan, Bringing design to the privacy table, in *Proc. the 2019 CHI Conference on Human Factors in Computing Systems*, Glasgow, UK, 2019, pp. 1–17.
[39] A. Dunne and F. Raby, Towards a critical design: Consuming monsters: Big, perfect, infectious, http://dunneandraby.co.uk/content/bydandr/42/0, 2005.
[40] S. Lawson, B. Kirman, C. Linehan, T. Feltwell, and L. Hopkins, Problematising upstream technology through speculative design, in *Proc. the 33rd Annual ACM Conference on Human Factors in Computing Systems*, Seoul, Republic of Korea, 2015, pp. 2663–2672.
[41] M. Malik and M. M. Malik, Critical technical awakenings, *Journal of Social Computing*, doi: 10.23919/JSC.2021.0035.
[42] N. JafariNaimi, L. Nathan, and I. Hargraves, Values as hypotheses: Design, inquiry, and the service of values, *Design Issues*, vol. 31, no. 4, pp. 91–104, 2015.
[43] C. A. L. Dantec and C. DiSalvo, Infrastructuring and the formation of publics in participatory design, *Social Studies of Science*, vol. 43, no. 2, pp. 241–264, 2013.
[44] K. M. MacQueen, N. T. Eley, M. Frick, L. R. Mingote, A. Chou, S. S. Seidel, S. Hannah, and C. Hamilton, Developing a framework for evaluating ethical outcomes of good participatory practices in TB clinical drug trials, *Journal of Empirical Research on Human Research Ethics*, vol. 11, no. 3, pp. 203–213, 2016.
[45] D. J. Mir, Y. Shvartzshnaider, and M. Latonero, It takes a village: A community based participatory framework for privacy design, presented at 2018 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW), London, UK, 2018.
[46] K. Shilton, Values levers: Building ethics into design, *Science*, *Technology*, & *Human Values*, vol. 38, no. 3, pp. 374–397, 2013.
[47] E. P. S. Baumer and M. S. Silberman, When the implication isnot to design (technology), in *Proc. the SIGCHI Conference on Human Factors in Computing Systems*, Vancouver, Canada, 2011, pp. 2271–2274.
[48] B. Friedman and D. G. Hendry, *Value Sensitive Design*. Cambridge, MA, USA: The MIT Press, 2019.
[49] D. D. Clark, *Designing an Internet*. Cambridge, MA, USA: The MIT Press, 2018.
[50] D. D. Clark, The contingent internet, *Daedalus*, vol. 145, no. 1, pp. 9–17, 2016.
[51] D. D. Clark, The design philosophy of the DARPA Internet Protocols, *ACM SIGCOMM Computer Communication Review*, vol. 25, no. 1, pp. 102–111, 1995.
[52] D. D. Clark, J. Wroclawski, K. R. Sollins, and R. Braden, Tussle in cyberspace: Defining tomorrow's Internet, *IEEE/ACM Transactions on Networking*, vol. 13, no. 3, pp. 462–475, 2005.
[53] S. Braman, The interpenetration of technical and legal decision-making for the internet, *Information*, *Communication* & *Society*, vol. 13, no. 3, pp. 309–324, 2010.
[54] H. Abelson, G. J. Sussman, *Structure and Interpretation of Computer Programs*. Cambridge, MA, USA: The MIT Press, 1996.
[55] E. Fisher, M. O'Rourke, R. Evans, E. B. Kennedy, M. E. Gorman, and T. P. Seager, Mapping the integrative field: Taking stock of socio-technical collaborations, *Journal of Responsible Innovation*, vol. 2, no. 1, pp. 39–61, 2015.
[56] A. S. Balmer, J. Calvert, C. Marris, S. Molyneux-Hodgson, E. Frow, M. Kearnes, K. Bulpin, P. Schyfter, A. Mackenzie, and P. Martin, Taking roles in interdisciplinary collaborations: Reflections on working in Post-ELSI spaces in the UK synthetic biology community, *Science and Technology Studies*, vol. 28, no. 3, pp. 3–25, 2015.
[57] F. Callard and D. Fitzgerald, *Rethinking Interdisciplinarity Across the Social Sciences and Neurosciences*. London, UK: Palgrave MacMillan, 2015.
[58] A. Quan-Haase, J. L. Suarez, and D. M. Brown, Collaborating, connecting, and clustering in the humanities, *American Behavioral Scientist*, vol. 59, no. 5, pp. 565–581, 2015.
[59] D. Fitzgerald and F. Callard, Social science and neuroscience beyond interdisciplinarity: Experimental entanglements, *Theory*, *Culture* & *Society*, vol. 32, no. 1, pp. 3–32, 2015.
[60] F. Callard, D. Fitzgerald, and A. Woords, Interdisciplinary collaboration in action: Tracking the signal, tracing the noise, *Palgrave Communications*, vol. 1, no. 1, p. 15019, 2015.
[61] A. Lucero, P. Dalsgaard, K. Halskov, and J. Buur, Designing with cards, in *Collaboration in Creative Design*, P. Markopoulos, J. -B. Martens, J. Malins, K. Coninx, and A. Liapis, eds. Cham, Switzerland: Springer International Publishing, 2016, pp. 75–95.
[62] B. Friedman and D. G. Hendry, The envisioning cards: A toolkit for catalyzing humanistic and technical imaginations, in *Proc. the SIGCHI Conference on Human Factors in Computing Systems*, Austin, TX, USA, 2012, pp. 1145–1148.
[63] A. Fedosov, M. Kitazaki, W. Odom, and M. Langheinrich, Sharing economy design cards, in *Proc. the 2019 CHI Conference on Human Factors in Computing Systems*, Glasgow, UK, 2019, pp. 1–14.
[64] J. Knapp, J. Zeratsky, and B. Kowitz, *Sprint: How to Solve Big Problems and Test New Ideas in Just Five Days*. New York, NY, USA: Simon and Schuster, 2016.
[65] S. Umbrello and O. Gambelin, Agile as a vehicle for values: A value sensitive design toolkit, doi: 10.13140/RG.2.2.17064.08965/1.
[66] C. Fiesler, Ethical considerations for research involving (speculative) public data, in *Proc. the ACM on Human-Computer Interaction*, vol. 3, no. GROUP, pp. 1–13, 2019.
[67] R. Y. Wong and V. Khovanskaya, Speculative design in HCI: From corporate imaginations to critical orientations, in *New Directions in Third Wave Human-Computer Interaction: Volume 2 – Methodologies*, M. Filimowicz and V. Tzankova, eds. Cham, Switzerland: Springer International Publishing, 2018, pp. 175–202.
[68] G. Cockton, Designing worth is worth designing, in *Proc.*

the 4th Nordic Conference on Human-Computer Interaction: Changing Roles, Oslo, Norway, 2006, pp. 165–174.

[69] G. Cockton, From doing to being: Bringing emotion into interaction, *Interacting with Computers*, vol. 14, no. 2, pp. 89–92, 2002.

[70] P. Sengers, K. Boehner, S. David, and J. "Jofish" Kaye, Reflective design, in *Proc. the 4th Decennial Conference on Critical Computing: Between Sense and Sensibility*, Aarhus, Denmark, 2005, pp. 49–58.

[71] C. DiSalvo, *Adversarial Design*. Cambridge, MA, USA: The MIT Press, 2012.

[72] M. Sloan, E. Moss, O. Awomolo, and L. Forlano, Participation is not a design fix for machine learning, arXiv preprint arXiv: 2007.02423, 2020.

[73] L. Irani, *Chasing Innovation: Making Entrepreneurial Citizens in Modern India*. Princeton, NJ, USA: Princeton University Press, 2019.

[74] L. Stark, D. Greene, and A. L. Hoffmann, Critical perspectives on governance mechanisms for AI/ML systems, in *The Cultural Life of Machine Learning*, J. Roberge and M. Castelle, eds. Cham, Switzerland: Palgrave Macmillan, 2021, pp. 257–280.



Luke Stark is an assistant professor in the Faculty of Information and Media Studies at Western University in London, ON, Canada. His work interrogating the historical, social, and ethical impacts of computing and AI technologies has appeared in journals including the *Information Society, Social Studies of Science, and New Media & Society*. He was previously a postdoctoral researcher in AI ethics at Microsoft Research, and a postdoctoral fellow in sociology at Dartmouth College; he received the PhD degree from Department of Media, Culture, and Communication at New York University in 2016, and the BA and MA degrees from University of Toronto in 2006 and 2008, respectively.